CCCA Cell-State Analysis — Karaayvas et al. 2018, Breast

Triple-Negative Breast Cancer · SmartSeq2 · 360 malignant cells (5 samples)

GO / NO-GO GATES

Gate 1 — mixing & continuity — GO

Harmony(theta=5) integrates all 5 patients: same-sample neighbour fraction 0.82 -> 0.28 (null 0.25), iLISI median 1.97 -> 4.12 of 5. One connected component (0 islands). Judged on the regenerated embedding + per-neighbourhood iLISI, not a scalar: samples intermix AND the manifold is continuous.

Gate 2 — module geography — GO

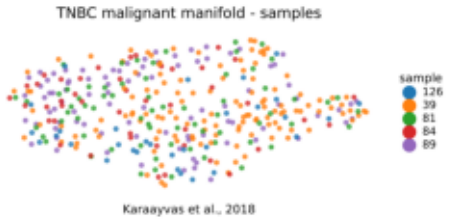
Non-branching case: a single pSom-origin -> terminus developmental gradient. pSom falls along pseudotime (rho -0.41); Gast & Crest both rise to the terminal tip (rho +0.79 / +0.82; Crest decile pt=0.98). joint pSom&Crest enrichment 0.37 (<1 = inverse terminal gradient, not a co-expression hub). FLAG (pass anyway): Crest co-enriches the Gast pole (smoothed Gast<->Crest r=+0.59), an inversion of the expected pSom-arm Crest — Gast, not pSom, tracks the neural-crest terminus here.

Gate 3 — cross-repression — NA

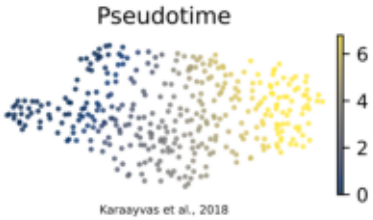
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks, fold-free). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks are not produced, per the deliverable spec).

STEP FIGURES

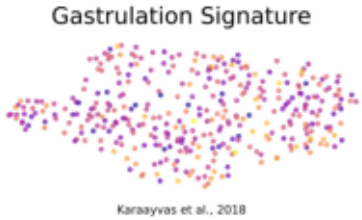
01 · Samples (Gate 1)



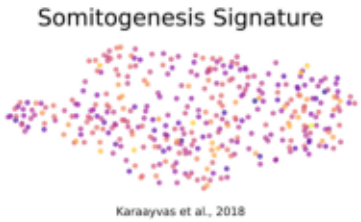
02 · Pseudotime



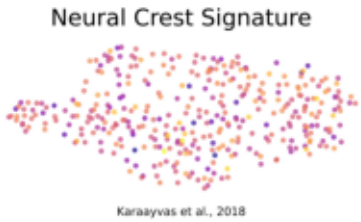
03 · Gastrulation



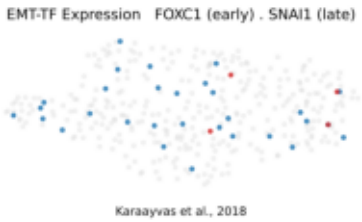
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs



Dataset: TNBC (Karaayvas 2018), SmartSeq2, 360 malignant cells over 5 patients; single low-cycling body (~14% cycling), cell-cycle-free axis (pseudotime vs cycling rho -0.19).

Verdict: Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous somitogenesis-origin -> neural-crest terminus axis. G1 GO / G2 GO / G3 NA, with the Crest-arm inversion flagged (Crest tracks the Gast terminus). EMT-TFs FOXC1 (early/root) -> SNAI1 (late).

PARAMETER CHOICES

Sample threshold	>=12 (geom-mean of dominant 31->5 spillover cliff, ratio 6.2) -> 5/6 samples, 360 cells
Overdispersed genes	51 (scFates find_overdispersed; sparse SmartSeq2)
Integration	Harmony on sample, theta=5 (mixing near null then flattens; avoids over-collapse)
Embedding	n_pcs=20, n_neighbors=30, UMAP spread=1.3; 1 component, 0 islands removed
Geometry	Non-branching -> scFates principal curve (ElPiGraph), Nodes=20, epg_lambda=0.3, epg_mu=100
Topology check	2 tips / 0 forks; fold self-approach 2.91 (>1 fold-free); continuity corr(x,t)=0.97
Root	tip 2 = Crest-low pSom-high developmental origin (derived from signature order)
Module means (raw)	Gast -0.220 · pSom +0.106 · Crest -0.067 (Linear-pSom/Crest)
EMT-TFs shown	FOXC1 (early/root, 33% expr) <-> SNAI1 (late/terminus); detection-gated, FOXC/PRRX in panel